

**OPC/MSEDE Tag Names & Communication****12/1/2022****Rev. 1****ABSTRACT:**

This document gives server and sub-server names and all Point-ID and Point Parameter information required to control the MUPP from third-party OPC-compliant software. This information can also enable MUPP control from Excel using MSEDE. Further, this document describes the process that must be carried out in the Experion Station HMI to allow third-party programs to gain control over all MUPP assets required to control the plant.

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Document Control and Revision History

Table 1: Document Control and Revision History

Revision	Date	Author(s)	Reviewed by	Comments/Details
0	2010	Earl Hopkinson		Table of parameters only
1	2022	Nicholas O’Keeffe		All sections new except for section 5 which is a modified version of rev0 document.

Definitions

AG	Agitator
BMP	Ball Mill Pump
CRP	Cyclone Recycle Pump
CUFP	Cyclone Underflow Pump
DT	Density Transmitter
EPKS	Experion Process Knowledge System
FCV	Flow Control Valve
FDP	Feed Disturbance Pump
FP	Feed Pump
FT	Flow Transmitter
HMI	Human Machine Interface
LT	Level Transmitter
LUP	Lamella Underflow Pump
MSEDE	Microsoft Excel Data Exchange
MUPP	Murdoch University Pilot Plant
NUFP/NTP	Needle Tank Underflow Pump
OPC	Open Protocol Communications
PP	Product Pump
SV	Solenoid Valve
TT	Temperature Transmitter

1.0 Purpose

The purpose of this document is to define details on the parameters required for control of the MUPP via third-party software. It is intended that this document is used to assist users in setting up control strategies using Open Protocol Communications (OPC); as such, it is recommended for use with OPC-compliant software. However, this information can also enable plant control via Microsoft Excel Data Exchange (MSEDE). Several documents are recommended for additional reading in section 6.0, which will help contextualise the information contained in this document, and further clarify the use of the MUPP.

2.0 Scope

This document gives server and sub-server names and all Point-ID and Point Parameter information required to control the MUPP from third-party OPC-compliant software. The Point-ID and Point Parameter information can also enable control using MSEDE. Further, this document describes the process that must be carried out in the Experion Station HMI to allow third-party programs to gain control over the assets described in the following section.

3.0 Tables of Parameters

The proceeding tables show the configuration details of each relevant parameter that can be read-from/written-to using the OPC protocol. Where changes have been enacted since the last revision, the assets and fields in question will be shown in red for clarity. Each table shows the asset in the first column (e.g., a pump, valve or transmitter), the point ID in the second column (synonymous with control modules in EPKS), and the point parameter in the third column. Only the server details (Table 2), Point ID, and Point Parameter (Columns 2 and 3 of Tables 3-11) are required to configure communication via MSEDSE or OPC.

Table 2: Pilot Plant Server Details.

SERVER NAME:	PPServer1
EPKS SUB-SERVER:	HWHsc.OPCServer

NOTE: Sub-server details NOT required for MSEDSE, only OPC applications.

Table 3: Level transmitters with associated point ID and Parameter.

<i>Asset:</i>	<i>Point ID:</i>	<i>Point Parameter:</i>
LT - 102	LT_102	LT_102.PV
LT - 122	LT_122	LT_122.PV
LT - 222	LT_222	LT_222.PV
LT - 322	LT_322	LT_322.PV
LT - 501	LT_501	LT_501.PV
LT - 542	LT_542	LT_542.PV
LT - 667	LT_667	LT_667.PV

Table 4: Flow transmitters with associated point ID and Parameter.

<i>Asset:</i>	<i>Point ID:</i>	<i>Point Parameter:</i>
FT - 148	FT_148	FT_148.PV
FT - 247	FT_247	FT_247.PV
FT - 347	FT_347	FT_347.PV
FT - 363	FT_363	FT_363.PV
FT - 401	FT_401	FT_401.PV
FT - 523	FT_523	FT_523.PV
FT - 569	FT_569	FT_569.PV
FT - 573	FT_573	FT_573.PV
FT - 613	FT_613	FT_613.PV
FT - 687	FT_687	FT_687.PV
FT - 689	FT_689	FT_689.PV

Table 5: Temperature transmitters with associated point ID and Parameter.

<i>Asset:</i>	<i>Point ID:</i>	<i>Point Parameter:</i>
TT - 568	TT_568	TT_568.PV
TT - 623	TT_623	TT_623.PV
TT - 643	TT_643	TT_643.PV
TT - 663	TT_663	TT_663.PV

Table 6: Density transmitters with associated point ID and Parameter.

<i>Asset:</i>	<i>Point ID:</i>	<i>Point Parameter:</i>
DT - 103	DT_103	DT_103.PV
DT - 123	DT_123	DT_123.PV
DT - 223	DT_223	DT_223.PV
DT - 323	DT_323	DT_323.PV

Table 7: Pump speed control with associated point ID and Parameter. *NOTE: RED TEXT INDICATES A CHANGE SINCE THE LAST REVISION.*

<i>Asset:</i>	<i>Point ID:</i>	<i>Point Parameter:</i>
BMP	BMP_REF_241-R1A	BMP241_OPC.PV
CRP	CRP_REF_341-R1A	CRP341_OPC.PV
CUFP	CUFP_REF_361-R1A	CUFP361_OPC.PV
FDP	FDP_REF_521-R1A	FDP521_OPC.PV
FP	FP_REF_141-R1A	FP141_OPC.PV
LUP	LUP_REF_421-R1A	LUP421_OPC.PV
NUFP	NTP_REF_561-R1A	NTP561_OPC.PV
PP	PP_REF_681-R1A	PP681_OPC.PV

NOTE: See section 4.0 *Enabling External Control in Station* for details pertaining to using FCVs in third-party programs.

Table 8: Pump state (ON/OFF) control with associated point ID and Parameter.

Asset	Point ID:	Point Parameter
BMP	BMP_OFF_241	BMP_ON_OFF.PVFL
CRP	CRP_OFF_341	CRP_ON_OFF.PVFL
CUFP	CUFP_OFF_361	CUFP_ON_OFF.PVFL
FDP	FDP_OFF_521	FDP_ON_OFF.PVFL
FP	FP_OFF_141	FP_ON_OFF.PVFL
LUFP	LUP_OFF_421	LUP_ON_OFF.PVFL
NUFP	NTP_OFF_561	NTP_ON_OFF.PVFL
PP	PP_OFF_681	PP_ON_OFF.PVFL

Table 9: Flow Control Valve position control with associated point ID and Parameter. *NOTE: RED TEXT INDICATES A CHANGE SINCE THE LAST REVISION.*

Asset	Point ID:	Point Parameter
FCV - 541	FCV_541-R1	FCV541_OPC.PV
FCV - 570	FCV_570-R1	FCV570_OPC.PV
FCV – 574 *OOC*	FCV_574-R1	FCV574_OPC.PV
FCV – 622 *OOC*	FCV_622-R1	FCV622_OPC.PV
FCV - 642	FCV_642-R1	FCV642_OPC.PV
FCV - 662	FCV_662-R1	FCV662_OPC.PV
FCV - 688	FCV_688-R1	FCV688_OPC.PV
FCV - 690	FCV_690-R1	FCV690_OPC.PV

NOTE: See section 4.0 *Enabling External Control in Station* for details pertaining to using FCVs in third-party programs.

Table 10: Agitator state (ON/OFF) control with associated point ID and Parameter.

Asset	Point ID:	Point Parameter
AG - 101	AG_101	AG_ON_OFF.PVFL
AG - 121	AG_OFF_121	AG_ON_OFF.PVFL
AG - 221	AG_OFF_221	AG_ON_OFF.PVFL
AG - 321	AG_321	AG_ON_OFF.PVFL
AG - 621	AG_621	AG_ON_OFF.PVFL
AG - 641	AG_641	AG_ON_OFF.PVFL
AG - 661	AG_661	AG_ON_OFF.PVFL

NOTE: To use control AGs over EXCEL or OPC, set AGs to program mode in Experion station

Table 11: Solenoid Valve position (OPEN/CLOSED) control with associated point ID and Parameter.

Asset	Point ID:	Point Parameter
SV - 104	SV_104	B13_4_1.PVFL
SV - 105	SV_105	B13_4_2.PVFL
SV - 124	SV_124	B13_4_4.PVFL
SV - 125	SV_125	B13_4_5.PVFL
SV - 614	SV_614	B13_5_10.PVFL
SV - 615	SV_615	B13_5_11.PVFL
SV - 616	SV_616	B13_5_12.PVFL
SV - 691	SV_691	B13_6_1.PVFL
SV - 692	SV_692	B13_6_2.PVFL
SV - 693	SV_693	B13_6_3.PVFL

NOTE: To use control SVs over EXCEL or OPC, set SVs to program mode in Experion station

Table 12: Controlled assets OPC routing status associated point ID and Parameter.

Asset	Point ID:	Point Parameter
BMP	BMP_REF_241-R1A	OPCenable.PVFL
CRP	CRP_REF_341-R1A	OPCenable.PVFL
CUFP	CUFP_REF_361-R1A	OPCenable.PVFL
FDP	FDP_REF_521-R1A	OPCenable.PVFL
FP	FP_REF_141-R1A	OPCenable.PVFL
LUFP	LUP_REF_421-R1A	OPCenable.PVFL
NUFP	NTP_REF_561-R1A	OPCenable.PVFL
PP	PP_REF_681-R1A	OPCenable.PVFL
FCV - 541	FCV_541-R1	OPCenable.PVFL
FCV - 570	FCV_570-R1	OPCenable.PVFL
FCV - 574 *OOC*	FCV_574-R1	OPCenable.PVFL
FCV - 622 *OOC*	FCV_622-R1	OPCenable.PVFL
FCV - 642	FCV_642-R1	OPCenable.PVFL
FCV - 662	FCV_662-R1	OPCenable.PVFL
FCV - 688	FCV_688-R1	OPCenable.PVFL
FCV - 690	FCV_690-R1	OPCenable.PVFL

4.0 Enabling External Control in Station

For safety purposes, all externally controlled flow controls and pumps (Table 7, Table 9) must be enabled from the EPKS HMI before the program will route externally controlled parameters to the control element. External control of each asset is enabled on a per-controller basis, the process of which is as follows:

1. Open the control parameters popup for the relevant plant asset in the MUPP HMI.
2. Check the 'Enable OPC Outer Loop' tick box (Figure 1).
3. Check the 'Controller Active' tick box (Figure 1).

NOTE: When controlling the MUPP via OPC it is worthwhile monitoring the status of OPC in EPKS, Table 12 shows the OPC enable status of each of the controlled assets; this parameter is a Boolean value that reads TRUE when control by third-party programs via OPC is being allowed.

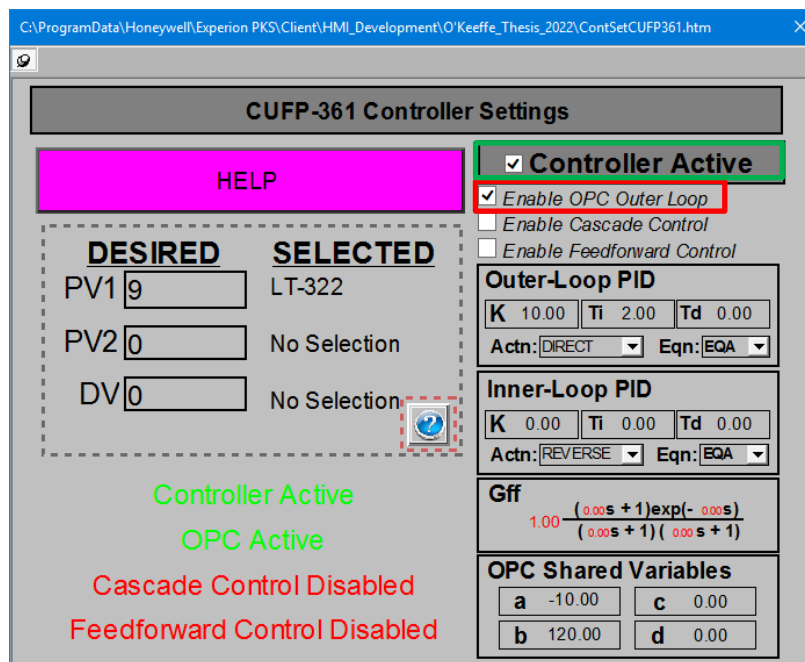


Figure 1: Cyclone Underflow Pump popup menu. The red box indicates OPC enable check box, and the green box indicates the controller enable checkbox. Both OPC and Controller enable checkboxes must be checked to allow for MUPP control from a third-party program.

5.0 Additional Reading

Consult Table 13 for recommendations for further reading.

Table 13: Recommended further reading and their relevance in the context of this document.

<i>Document:</i>	<i>Relevance:</i>
PP-5001: Operating Pilot Plant via Excel	PP-5001 outlines how to implement parameter control from MATLAB using Excel and Microsoft Excel Data Exchange.
PP-5002: Operating Pilot Plant via Station Rev. 1	PP-5002 gives insight into the operation of the MUPP in Experion Station, including guides on controller configuration.
PP-5007: 'OPCControlTemplateR0': Use and Function Rev. 0	PP-5007 outlines how to implement parameter control from MATLAB using the 'OPCControlTemplateR0' Program.